

RETAIL FOOD STORES ANALYTICS DASHBOARD

Submitted in partial fulfillment of the requirements for the award of degree of

**Bachelor of Technology
(Computer Science and Engineering)**

**Submitted to
LOVELY PROFESSIONAL UNIVERSITY
PHAGWARA, PUNJAB**



SUBMITTED BY

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CERTIFICATE

This is to certify that **Prema Sai Kimmi** bearing Registration no. **12306000** has completed **INT 374** project titled, “Retail food store analytics and Dashboard Reporting Using Power BI” under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

School of Computer Science and Engineering
Lovely Professional University
Phagwara, Punjab.

DECLARATION

I **Prema Sai Kimmi**, student of **B. Tech CSE - K23CB** under CSE Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Signature : Prema Sai Kimmi

Registration No. **12306000**

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who contributed to the successful completion of this report. I am thankful to my faculty and mentors for their guidance, valuable suggestions, and continuous support throughout this project. Their insights helped me understand the concepts clearly and apply them effectively.

I would also like to acknowledge the resources and tools that supported this work, especially the use of data analysis and visualization techniques, which played a key role in developing the dashboard and deriving meaningful insights. Lastly, I am grateful to my friends and peers for their encouragement and support during the completion of this report.

1. Introduction

In today's competitive and data-driven business environment, understanding the distribution and operational characteristics of retail food stores is essential for effective planning, regulation, and business expansion. Retail food stores play a crucial role in urban development, consumer access to food, and economic activity. This project focuses on the development of an **interactive Power BI dashboard** that analyzes **retail food store distribution, establishment patterns, and store size characteristics** across multiple counties, cities, and streets. The dashboard transforms raw licensing and store data into **clear, visual, and actionable insights**, enabling users to identify high-density retail zones, dominant establishment types, and variations in store size. The primary goal of this dashboard is to support data-driven decision-making for stakeholders such as business analysts, urban planners, and regulatory authorities.

2. Dataset Description

The dataset used in this project consists of **retail food store license data**, containing detailed information about store locations, operational characteristics, and physical size.

The dataset includes the following attributes:

- Geographic details: State, County, City, Zip Code
- Location details: Street Name, Street Number
- Store identification: License Number, DBA Name, Entity Name
- Operational attributes: Establishment Type, Operation Type
- Store size metrics: Square Footage, Total Store Area

This dataset enables **geographic, operational, and size-based analysis** of retail food stores at multiple levels such as county, city, street, and establishment type.

3. Objectives of the Dashboard

The main objectives of this dashboard are:

- To analyze the geographic distribution of retail food stores across counties and cities.
- To identify high-density retail areas at city and street levels
- To analyze store size patterns and total retail area distribution
- To understand the dominance of different establishment types
- To enable interactive and user-friendly exploration of retail data through Power BI

4. Dashboard Design Overview

The dashboard is designed as a **multi-page Power BI report** with a clear separation between navigation, high-level insights, and detailed analysis. Navigation buttons and tooltips are used to enhance usability and maintain a clean layout.

Navigation Features

- Dedicated **Home page** for navigation and orientation
- Clickable buttons for smooth page transitions
- Slicers for county and establishment type filtering
- Tooltip pages for additional contextual insights

This structure ensures a **professional, user-friendly, and scalable dashboard design**.

5. Dashboard Pages Description

5.1 Home Page – Navigation Interface

The Home page acts as a **landing page** for the dashboard. It provides:

- Dashboard title and analytical scope
- Navigation buttons to Overview, Store Insights, and Tooltip pages

The Home page is intentionally kept free of charts to **avoid distraction** and improve user experience by separating navigation from analysis.

5.2 Overview Dashboard

This page provides a **high-level summary** of retail food store distribution.

Key KPIs Displayed

- Total Stores
- Total Store Area
- Average Store Size

- Maximum Store Size
- Count of Cities
- Count of Counties

Visuals Used

- KPI cards for summary metrics
- Map visualization for county-level distribution
- Bar charts for top cities and top streets by store count
- Bar and donut charts for establishment type distribution
- Treemap for county–city hierarchical analysis

Purpose

To provide users with a **quick and comprehensive understanding** of where retail food stores are located and which regions dominate retail activity.

5.3 Store Insights Dashboard

This page focuses on **store size and operational characteristics**.

Visuals Used

- KPI cards for store size metrics
- Bar chart for average store size by establishment type
- Table for top-ranked streets with total store area
- Bar chart for top cities by number of retail food stores

Purpose

To analyze **store capacity, operational scale, and size variation**, supporting deeper insights into retail infrastructure and business intensity.

5.4 Tooltip Pages (Street Tooltip & General Tooltip)

Tooltip pages are designed to appear when users **hover over visuals**.

Purpose of Tooltips

- Provide additional insights without cluttering main pages
- Improve dashboard readability and interaction
- Enable contextual, on-demand analysis

Tooltips enhance **analytical depth while maintaining visual simplicity**.

6. Key Measures and Calculations

To support accurate and dynamic analysis, the following measures were created using DAX:

- Total Stores
- Total Store Area
- Average Store Size
- Maximum Store Size
- Store Count by City
- Store Count by Street
- Store Distribution by Establishment Type

These measures ensure **real-time interaction, dynamic filtering, and consistent calculations** across all visuals.

7. Tools and Technologies Used

- Microsoft Power BI Desktop
- DAX (Data Analysis Expressions)
- Data Modeling and Relationships
- Interactive Visuals (Charts, Maps, Tables)
- Navigation Buttons, Slicers, and Tooltips

8. Conclusion

The Retail Food Stores Analytics Dashboard effectively transforms raw retail license data into meaningful and actionable insights. By combining geographic analysis, store size evaluation, and establishment-type comparison, the dashboard supports **data-driven planning and decision-making**.

This project demonstrates practical skills in:

- Data visualization and storytelling
- Power BI dashboard design
- KPI development and interactivity
- Business and urban retail analysis

The dashboard can be further enhanced by adding **time-based trends, predictive analytics, and revenue estimation models**, making it even more valuable for strategic planning.

LinkedIn : https://www.linkedin.com/posts/prema-sai_powerbi-dataanalytics-dashboarddesign-activity-7406744758070468608-VEENV/?utm_source=share&utm_medium=member_desktop&rcm=ACoAAEcgeHQBxGnHuXFDPI9H0EikSzmR_lu_qQc

Dataset : <https://catalog.data.gov/dataset/retail-food-stores>

